Auditing Emotion Recognition Systems for Bias Against Neurodiverse Individuals

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Abstract

Facial Emotion Recognition (FER) systems are increasingly deployed in high-stakes environments such as hiring and education, yet they are predominantly trained on neurotypical data, raising significant ethical concerns regarding fairness and inclusivity. This study audits two prominent open-source FER systems—DeepFace, a widely adopted industry standard, and EmotiEffLib, a state-of-the-art efficient model—to evaluate their performance and calibration when applied to neurodivergent populations. We assessed model behavior across three standard "in-the-wild" benchmarks (RAF-DB, AffectNet, FER-2013) and a specific test set of 220 images of autistic children (FER-Autism).Our quantitative analysis reveals a systemic "collapse to neutral" bias, where models misinterpret the "flat affect" often associated with neurodivergence as emotional neutrality. While EmotiEffLib achieves high accuracy on standard benchmarks (up to 73%), its performance drops by nearly half (to 37%) on the autism dataset. Crucially, we find that EmotiEffLib loses its calibration safeguards on neurodivergent data, shifting from "knowing when it is unsure" to being "confidently wrong." DeepFace exhibits severe overconfidence ($ECE > 0.4$) across all domains, a failure we trace to its reliance on the noisy FER-2013 training set. Qualitative analysis using Grad-CAM and LIME confirms that DeepFace relies on rigid, upper-face heuristics, whereas EmotiEffLib utilizes more distributed cues but still fails to capture atypical expressive patterns. Based on these findings, we propose a "Fairness Auditing Checklist" that operationalizes ethical principles into technical safeguards, advocating for calibration-aware audits to prevent algorithmic ableism in emotion AI.

Contents

1 Introduction

1.1 Background on Emotion Recognition (ER) Systems

Emotion recognition (ER) systems are a class of affective computing technologies designed to infer human emotional states from observable signals such as facial expressions, speech, posture, or physiological responses. Among these approaches, facial emotion recognition (FER) has become one of the most widely studied and deployed, driven by advances in computer vision and deep learning as well as the availability of large-scale annotated facial image datasets [17, 35].

Most contemporary FER systems rely on supervised machine learning, particularly convolutional neural networks (CNNs) and related deep learning architectures. These models are typically trained on benchmark datasets such as FER-2013, AffectNet, RAF-DB, and CK+, which label facial images according to discrete emotion categories derived from psychological theories of basic emotions [17]. Model performance is commonly evaluated using metrics such as accuracy, F1-score, or top-k accuracy on held-out test sets drawn from similar distributions.

With improvements in computational efficiency and predictive performance, FER technologies have increasingly been integrated into real-world applications. These include automated video interviews and personality assessment in recruitment, engagement and attention monitoring in educational environments, and emotion-aware systems in health-care and assistive technologies [41, 4]. In such contexts, FER outputs may influence judgments about engagement, competence, emotional appropriateness, or well-being.

Despite their growing adoption, ER systems are often developed and validated under controlled conditions that prioritize benchmark performance. This has raised concerns regarding their robustness, interpretability, and generalizability when deployed in socially complex environments involving diverse users, particularly those whose expressive patterns diverge from the assumptions embedded in dominant emotion recognition models [6].

1.2 Problem Statement: Bias Against Neurodiverse Individuals

A central challenge facing facial emotion recognition systems is their systematic misalignment with neurodivergent forms of emotional expression. Most FER models are trained on datasets dominated by neurotypical faces and annotated according to normative assumptions about how emotions are expressed and perceived. As a result, these systems implicitly treat neurotypical expressivity as the default standard, while deviations from this norm are more likely to be misclassified as neutral, ambiguous, or incorrect [29, 33].

Neurodivergent individuals, including those on the autism spectrum and individuals with intellectual or developmental differences, often exhibit distinct expressive, perceptual, and attentional patterns. These may include reduced facial expressivity, atypical gaze behavior, differences in timing or intensity of expressions, and a greater reliance on contextual cues rather than facial signals alone. When FER systems are applied to such populations, these differences frequently lead to systematic generalization failures that are not attributable to emotional deficit but to representational mismatch between model assumptions and human variability [29, 57].

This mismatch is reinforced by upstream design choices in data collection and annotation. Emotion datasets commonly exhibit low inter-rater reliability and limited representation of neurodivergent participants, resulting in unstable or exclusionary ground truth labels [6]. Consequently, FER models trained on these datasets inherit epistemic biases that affect both their predictive behavior and their confidence estimates, increasing the likelihood of disproportionate misclassification for neurodivergent users.

The implications of such misclassification extend beyond technical performance. In high-stakes contexts such as hiring, education, and healthcare, FER outputs may influence decisions related to suitability, engagement, emotional appropriateness, or mental state. Empirical studies indicate that emotion AI systems deployed in these settings are frequently experienced as unfair, exclusionary, or stigmatizing, particularly by neurodivergent individuals [19, 4, 41]. These risks raise significant ethical concerns regarding fairness, accountability, and inclusivity in emotion AI.

Together, these challenges motivate the need for systematic investigation into how FER systems behave when applied to neurodivergent populations. Rather than proposing new predictive models, this project addresses the problem by auditing existing FER systems to examine their generalization behavior, attention patterns, and decision logic using explainability methods. By grounding this audit in established ethical AI principles, the project aims to assess whether current emotion recognition technologies operate accurately, fairly, and ethically in contexts involving neurodiverse users [50, 43].

1.3 Research Questions and Project Goals

The primary goal of this project is to investigate whether facial emotion recognition (FER) systems operate accurately, fairly, and ethically when applied to neurodivergent individuals. While emotion recognition technologies are increasingly deployed in real-world settings, many systems are developed and evaluated using datasets that primarily reflect neurotypical patterns of facial expressivity. This raises concerns about their ability to generalize to individuals whose emotional expressions may differ from these normative assumptions, as well as the potential ethical harms that may arise from misclassification in high-stakes decision-making contexts.

To address these concerns, this project is guided by the following research questions:

- **RQ1:** How accurately and fairly do facial emotion recognition systems classify expressions of neurodivergent individuals, and what implications do atypical expressive patterns have for model generalization and bias?

This question focuses on evaluating the performance and fairness of emotion recognition models when confronted with non-neurotypical facial expressions, with particular attention to generalization failures and systematic biases.

- **RQ2:** What ethical harms emerge from the misclassification of neurodivergent individuals by emotion recognition systems, particularly in decision-making contexts such as hiring and education?

  This question examines the potential social and ethical consequences of misclassification, including risks of exclusion, discrimination, and stigmatization in sensitive application domains.

- **RQ3:** How have ethical AI frameworks, such as the EU HLEG Guidelines for Trustworthy AI, been applied to support fairness audits and the inclusive design of facial emotion recognition systems for neurodiverse populations?

  This question explores how established ethical AI principles can be used to guide the auditing and evaluation of emotion recognition technologies, with the aim of supporting more inclusive and responsible system design.

2 Literature Review

This section synthesizes existing research on facial emotion recognition systems with a focus on bias, generalization failures, and ethical risks affecting neurodivergent individuals. The review is organized thematically, examining technical limitations of FER models, evidence on neurodivergent expressivity, and the implications of deploying emotion recognition technologies in high-stakes contexts. Together, these strands establish the empirical and conceptual foundation for the fairness auditing approach adopted in this project.

2.1 Bias and Generalization Failures in Facial Emotion Recognition

A substantial body of research demonstrates that affective computing systems, particularly facial emotion recognition (FER) models, exhibit persistent bias, limited generalization, and reliability concerns when applied outside the narrow conditions under which they are typically developed. These shortcomings are not isolated implementation errors but reflect structural assumptions embedded in dataset construction, labeling practices, and dominant emotion modeling paradigms.

Early empirical work revealed that FER classifiers trained predominantly on neurotypical facial data systematically misclassified autistic children's expressions, resulting in accuracy disparities of 15–20% [29]. Subsequent studies confirmed that such errors arise from generalization failures across identity and expressive variation: person-dependent models outperform cross-group models by approximately 9%, indicating that expressive individuality destabilizes learned emotion mappings [16]. While normalization approaches such as Deep Face Normalization attempt to decouple identity from expression, they do not address the deeper assumption that emotional expression is stable, uniform, and universally legible.

These generalization failures are compounded by instability in emotion ground truth labels. Analyses of benchmark datasets such as AffectNet and EMOTIC report consistently low inter- and intra-rater reliability, with Krippendorff's α values frequently below 0.7, calling into question the objectivity of emotion annotations [6]. Comparable issues arise in commercial systems, where inter-system agreement approaches randomness (Cohen's $\kappa \approx 0.06$) and performance varies substantially across demographic groups [55]. Such disparities, often favoring Western, younger, and lighter-skinned faces, suggest that algorithmic discrimination is primarily driven by data imbalance and labeling subjectivity rather than inherent physiological differences [31].

Technical evaluations further illustrate a disconnect between benchmark performance and real-world robustness. Although convolutional neural network (CNN)–based FER models achieve high accuracy on posed datasets, real-world accuracy often declines by approximately 20% due to reliance on exaggerated expressions and culturally homogeneous training data [17]. Commercial systems including Microsoft Azure, Face++, and FaceReader routinely exceed 95% accuracy on curated datasets yet fall below 60% when applied to spontaneous expressions, disproportionately labeling subtle affect as "neutral" [28]. Even advanced attentional architectures continue to rely on Ekman's discrete emotion taxonomy, privileging highly expressive categories such as happiness and surprise while marginalizing low-intensity or ambiguous emotions [35].

Autism-focused research underscores the ethical salience of these technical limitations. Multiple studies show that commercial FER systems fail to generalize to neurodivergent populations despite evidence that autistic and non-autistic children do not differ in inherent emotional capacity [33]. Retraining models on heterogeneous datasets improves both accuracy and fairness by up to 20%, highlighting the central role of representational diversity rather than architectural sophistication alone [22]. Fairness-oriented mitigation strategies, including demographic bias mitigation frameworks and domain adaptation techniques, reduce some disparities [49], yet they do not resolve the epistemic constraints imposed by universalist emotion models.

Architectural differences further shape these outcomes. CNNs trained on neurotypical facial data consistently misclassify individuals with intellectual disabilities, while transformer-based models demonstrate improved overall performance but continue to struggle with subtle or low-intensity emotions [43, 42]. These persistent gaps suggest that bias cannot be eliminated through optimization alone. As Devillers et al. (2023) and Kumar et al. (2024) argue, affective computing remains constrained by universalist assumptions that privilege uniformity over variability [10, 27]. Similar normative biases appear in psychometric instruments, where revisions to the Benton Facial Recognition Test (BFRT-r) reveal compensatory perceptual strategies that traditional metrics misclassify, paralleling failures observed in FER systems [38].

Alongside these empirical findings, critical scholarship challenges the universality hypothesis underpinning emotion AI, questioning the assumption that facial expressions reliably map onto internal emotional states [34]. Ethical analyses introduce the concept of *affective indiscrimination* to describe the harm caused when contextually appropriate affective behavior is misinterpreted as deviant [8]. Empirical evidence from hiring contexts supports these concerns, with job applicants frequently reporting emotion AI evaluations as unjust and dehumanizing, particularly among neurodivergent, transgender, and racialized groups [19]. While these critiques extend beyond technical performance, they reinforce the need to interrogate how emotion AI systems operationalize normative assumptions.

Taken together, this literature establishes that bias in FER systems arises from structural assumptions about emotional universality, unstable labeling practices, and unrepresentative training data. These limitations motivate a shift from performance-centric evaluation toward systematic auditing of generalization behavior, particularly for neurodivergent expression profiles.

2.2 Neurodivergent Expressivity and the Limits of Emotion Models

Research on neurodivergent expressivity further exposes the fragility of FER systems grounded in discrete and neurotypical emotion models. Empirical evidence consistently shows that individuals on the autism spectrum and related neurodivergent groups convey emotion through subtler, context-dependent cues that resist standardization.

Studies of automatic emotion recognition in clinical and assistive settings indicate that pre-trained commercial models trained exclusively on neurotypical samples underperform when applied to neurodivergent users [40]. Reviews of intelligent technologies for autism-related applications reveal pervasive methodological constraints, including small sample sizes, Western-centric datasets, and limited attention to fairness or inclusivity [26]. Although some hybrid CNN ensembles demonstrate high accuracy for early autism detection [9], these systems remain constrained by homogeneous data and limited generalization.

Behavioral and perceptual research clarifies the sources of this mismatch. Neurodivergent individuals often exhibit reduced activation of facial action units, leading FER systems to misclassify valid emotional expressions as neutral or ambiguous [29]. Eye-tracking studies demonstrate distinctive gaze strategies among individuals with autism and schizophrenia, characterized by reduced fixation on socially normative facial regions [20]. When these perceptual differences are explicitly modeled, recognition accuracy improves, indicating that misclassification reflects model assumptions rather than expressive deficits. Complementary findings suggest that challenges in emotion recognition may arise from contextual reasoning demands rather than facial decoding alone [39]. High intra-group variability further complicates generalization, with participant-independent models yielding low AUC scores due to divergent perceptual strategies [3, 52].

Meta-analytic evidence supports these conclusions. A synthesis of 148 studies reports robust, nonselective impairments across conventional FER paradigms, underscoring the inadequacy of universal emotion models for capturing neurodivergent expressivity [57]. In response, recent work increasingly emphasizes participatory and neuroinclusive design approaches. Frameworks such as the Social–Emotional–Sensory Design Map center embodied and relational emotion models [59], while participatory telehealth interventions promote emotional self-determination rather than normalization [53]. Temporal affect tracking further challenges stereotypes of emotional unpredictability by revealing structured emotional variability in neurodivergent populations [18].

Technological innovations, including thermal imaging and multimodal FER systems, demonstrate potential for mitigating visual-normative bias [24, 32, 1]. GAN-based facial reenactment approaches similarly aim to support assistive recognition without pathologizing difference [23]. However, these advances primarily illustrate the limitations of unidimensional emotion models rather than offering immediate solutions for deployed FER systems.

At a theoretical level, critiques of design ideals centered on "humanness" and cognitive normativity advocate epistemic humility and participatory inclusion as ethical imperatives [45, 2]. Empirical reviews nevertheless show that neurodivergent participation remains rare in inclusive AI initiatives [56]. Collectively, this literature reframes neurodivergent expressivity not as error or deficit but as evidence of affective pluralism, exposing the limits of universalist emotion recognition frameworks.

2.3 Ethical Risks and the Case for Fairness Auditing

The deployment of emotion recognition systems in recruitment, education, and healthcare amplifies existing structural inequities by transforming affect into a metric of surveillance, evaluation, and control. Ethical frameworks emphasize transparency, consent, calibration, and recourse as safeguards against such harms [15], yet empirical evidence suggests that these protections are often insufficient in practice.

Studies of AI-mediated interviews indicate that automated affective assessments are widely perceived as unfair, opaque, and invasive [41]. In educational contexts, FER-based engagement monitoring risks penalizing autistic students by equating attentiveness with neurotypical facial expressivity [4]. Healthcare applications raise concerns of diagnostic overreach and privacy, particularly when emotion inference substitutes for contextual clinical judgment [40, 21]. Across these domains, the illusion of consent in institutional settings further compounds ethical risk [7, 5].

Technical fairness-by-design interventions provide partial mitigation. Indirect adversarial learning approaches demonstrate that removing sensitive demographic cues can reduce bias without explicit labeling [14]. However, scholars caution that such methods do not address deeper epistemic assumptions that conflate emotional legitimacy with detectability. Critical accounts describe this dynamic as *affective surveillance*, in which emotion AI enforces conformity through automated interpretation [19]. Normative frameworks of affective justice argue that ethical emotion recognition should respect contextual appropriateness rather than prioritize classification accuracy alone [8].

Governance frameworks increasingly categorize FER as a high-risk technology, calling for explainability, auditing, and accountability mechanisms [36, 6, 50]. Explainable and human-in-the-loop approaches offer practical pathways for operationalizing these principles, enabling the inspection of attention patterns, confidence calibration, and failure modes [43, 22, 11]. Such diagnostic tools support fairness auditing practices that move beyond benchmark accuracy toward context-sensitive evaluation.

Across these domains, a consistent theme emerges: affective computing systems inherit technical and epistemic biases that disproportionately marginalize neurodivergent expression. Achieving inclusivity therefore requires a dual strategy—empirical auditing of algorithmic behavior and a reconceptualization of emotion as plural, contextual, and embodied. This framework underpins the rationale for the fairness-aware, neurodiversity-informed auditing protocol proposed in this project.

Synthesis Summary. Across the reviewed corpus, three overarching insights emerge. First, bias and generalization failures in emotion AI are structurally embedded in data practices and universalist modeling assumptions. Second, evidence on neurodivergent expressivity demonstrates that emotion is plural and context-dependent, rendering one-size-fits-all recognition paradigms inadequate. Third, ethical governance and affective justice frameworks reveal that fairness extends beyond algorithmic design to the institutional contexts in which emotion AI operates. Together, these insights motivate a shift from affective computation toward affective ethics, where legitimacy depends not on detecting emotion, but on understanding and respecting expressive diversity.

2.4 Overview of Explainability Methods (Grad-CAM, Occlusion, and LIME)

Concerns regarding the reliability, validity, and ethical implications of facial emotion recognition (FER) systems have led to increasing calls for transparency and interpretability in affective computing research. A growing body of work argues that conventional performance metrics such as accuracy and F1-score are insufficient for evaluating the trustworthiness of emotion recognition systems, particularly when these systems are deployed in socially sensitive or high-stakes contexts [6, 34]. Explainable Artificial Intelligence (XAI) methods offer mechanisms to inspect how models arrive at their predictions, thereby enabling researchers to assess whether decisions are grounded in semantically meaningful facial cues or rely on spurious correlations.

Within affective computing, explainability is increasingly framed as a diagnostic tool rather than a post-hoc justification mechanism. This perspective is especially relevant when considering neurodiverse populations, where facial expressivity may deviate from neurotypical norms and challenge the assumptions implicitly encoded in training data and model architectures [27]. In this context, explainability methods support ethical auditing by revealing patterns of attention, feature dependence, and decision instability that may contribute to systematic bias or exclusion.

Gradient-based visual explanations. Gradient-based explainability methods aim to identify regions of the input image that contribute most strongly to a model's prediction. A widely adopted approach in computer vision is Gradient-weighted Class Activation Mapping (Grad-CAM), which produces coarse localization maps by backpropagating gradients from a target class to convolutional feature maps [47].

In facial emotion recognition, such visual explanations are commonly used to evaluate whether models attend to anatomically and psychologically meaningful facial regions, such as the eyes, mouth, and eyebrows, that are associated with emotional expression. Prior work has shown that attention-based visualizations can reveal emotion-specific facial cues and expose differences in how models process distinct emotional categories [35]. At the same time, critical analyses caution that visually plausible attention maps do not guarantee semantic correctness, particularly when models are trained on constrained or biased datasets [34]. Consequently, gradient-based explanations are best interpreted as indicative signals of model focus rather than definitive evidence of causal reasoning.

Occlusion sensitivity analysis. Occlusion sensitivity is a perturbation-based explainability technique that evaluates model robustness by systematically masking regions of the input image and observing the resulting changes in predictions. This approach was originally introduced as a means of visualizing and understanding convolutional neural networks by identifying regions whose removal most strongly affects model output [58]. Unlike gradient-based methods, occlusion sensitivity does not rely on internal model gradients and therefore provides a complementary, model-agnostic perspective.

In the context of emotion recognition, both computational and psychological studies indicate that specific facial regions play diagnostic roles for different emotions, and that reliance on these regions varies across individuals and contexts [25]. Occlusion analysis enables researchers to assess whether FER systems exhibit narrow dependence on specific facial cues, which may increase vulnerability to expressive variability, partial occlusion, or atypical expression patterns. Such brittleness is particularly relevant when auditing systems intended for use with neurodivergent individuals, whose facial expressions may not conform to dominant expressive norms.

Local surrogate explanations (LIME). Local Interpretable Model-agnostic Explanations (LIME) provide instance-level explanations by approximating a complex model's behavior locally using an interpretable surrogate model [44]. In image-based tasks, LIME operates by perturbing super-pixels and measuring their influence on the predicted class, thereby highlighting regions that are locally important for a specific prediction.

In facial emotion recognition research, LIME has been applied to examine whether model decisions are driven by coherent facial structures or by fragmented and potentially irrelevant regions. Prior studies using LIME have demonstrated that explanation patterns may differ substantially across populations, including between disabled and non-disabled individuals, raising concerns about stability and generalization [43]. Fragmented or peripheral explanations are often interpreted as indicators of unreliable or weakly grounded decision boundaries.

Explainability for fairness auditing. Recent ethical scholarship emphasizes that explainability should be integrated into broader evaluation frameworks that address fairness, accountability, and transparency, rather than treated as an isolated technical feature [27]. In affective computing, explainability methods have also been used to reveal substantial disagreement between different emotion recognition systems operating on the same inputs, underscoring the normative and contested nature of automated emotion inference [55].

Taken together, gradient-based visual explanations, occlusion sensitivity analysis, and local surrogate models such as LIME provide complementary perspectives on model behavior. Their combined use enables triangulation of attention localization, feature dependence, and decision stability, forming a principled foundation for auditing facial emotion recognition systems with respect to bias, robustness, and inclusivity. This triangulated explainability approach is particularly well suited for examining ethical risks posed to neurodivergent individuals, whose expressive diversity challenges the assumptions underlying many FER models.

3 Methodology

3.1 Literature Review Methodology

3.1.1 Databases and Search Strategy

To identify relevant studies, a systematic literature search was conducted across the following academic databases:

- Scopus
- IEEE Xplore
- PubMed

Search queries were constructed using combinations of keywords related to facial emotion recognition, neurodiversity, bias, and ethical considerations. Separate Boolean search strings were defined for each research question.

RQ1 Search String

```
("facial emotion recognition" OR "facial expression recognition" OR "affective comput
AND (autism OR "autism spectrum disorder" OR neurodivergent OR neurodiversity
OR "down syndrome" OR "atypical expression")
AND (accuracy OR performance OR classification)
AND (bias OR fairness OR generalization)
```

RQ2 Search String

```
("facial emotion recognition" OR "facial expression recognition" OR "affective comput
AND (autism OR "autism spectrum disorder" OR neurodivergent OR neurodiversity
OR "down syndrome" OR "atypical expression")
AND (bias OR misclassification OR discrimination OR exclusion OR harm)
AND (ethics OR "ethical implications" OR "algorithmic bias" OR "social impact" OR fai
AND ("hiring" OR employment OR workplace OR education OR learning OR school OR classr
```

RQ3 Search String

```
("facial emotion recognition" OR "facial expression recognition" OR "affective comput
AND (autism OR "autism spectrum disorder" OR neurodivergent OR neurodiversity
OR "down syndrome" OR "atypical expression")
```

```
AND ("AI ethics" OR "ethical AI" OR "responsible AI" OR "Trustworthy AI"
OR "AI governance" OR "AI fairness" OR "algorithmic fairness"
OR "fairness audit" OR "inclusive design" OR inclusivity)
```

3.1.2 Inclusion and Exclusion Criteria

To ensure relevance and quality, predefined inclusion and exclusion criteria were applied during the study selection process.

Inclusion Criteria

- Peer-reviewed journal articles, conference papers, or reputable preprints (e.g., arXiv) published between 2014 and 2025;
- Articles written in English;
- Empirical studies, systematic reviews, or case studies;
- Research addressing facial emotion recognition, facial expression analysis, or affective computing systems based on machine learning or deep learning methods;
- Studies involving or discussing neurodivergent populations, including individuals with autism spectrum disorder, Down syndrome, or atypical expressive patterns;
- Papers examining model performance, fairness, or ethical dimensions of emotion recognition, including misclassification harms and applications in education, hiring, or healthcare.

Exclusion Criteria

- Non–peer-reviewed works (e.g., opinion pieces, editorials, blogs, or reports);
- Studies focused solely on technical model optimization without discussion of bias, fairness, or ethics;
- Research limited to non-visual modalities such as speech or text-based emotion recognition;
- Studies centered on therapeutic or clinical interventions for neurodivergent individuals;
- Papers discussing general AI ethics without explicit reference to emotion recognition or neurodiversity;
- Publications written in languages other than English.

3.2 Screening Protocol

This study follows the PRISMA 2020 guidelines for conducting and reporting systematic literature reviews. A structured and reproducible screening process was employed to identify relevant studies on bias, fairness, and neurodiversity in facial emotion recognition systems.

3.2.1 Search and Deduplication

An initial literature search was conducted across multiple academic databases, including Scopus, IEEE Xplore, and PubMed, using predefined Boolean search strings aligned with the study's research questions. The search yielded a total of 3,475 records, which were imported into the Rayyan systematic review platform for screening and management.

Rayyan's automated duplicate detection flagged 896 records as potential duplicates. These records were manually reviewed to resolve false positives and clustered entries. Following manual verification, 497 records were removed as confirmed duplicates, while the remaining records were retained for subsequent screening. The resulting set of unique records was then advanced to the title and abstract screening stage.

3.2.2 Title and Abstract Screening

All remaining records were screened at the title and abstract level using predefined inclusion and exclusion criteria. Screening focused on relevance to affective computing and facial emotion recognition, explicit consideration of neurodivergent populations, and discussion of fairness or ethical implications.

Following this stage, 2,919 records were excluded. The primary reasons for exclusion included lack of relevance to emotion recognition, absence of neurodiversity-related analysis, and non-peer-reviewed publication status. The remaining records were retained for full-text eligibility assessment.

3.2.3 Eligibility and Final Inclusion

After resolving borderline cases through a full-text analysis of the 59 records, 50 studies were deemed eligible and included in the final literature corpus. These studies formed the basis for the qualitative thematic synthesis presented in the Related Work and Literature Review Findings sections. The final selection reflects a focused body of peer-reviewed research addressing bias, expressivity, and ethical risks in emotion recognition systems, particularly as they affect neurodivergent populations.

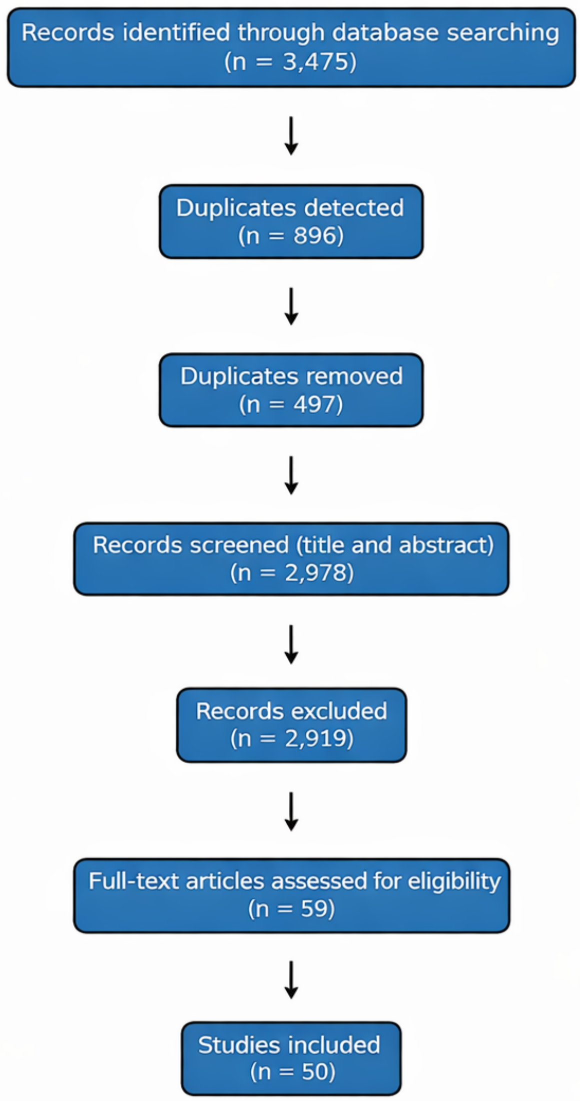

Figure 1: PRISMA flow diagram summarizing the identification, screening, eligibility, and inclusion stages.

3.3 Data and Models Under Evaluation

To empirically evaluate the performance of emotion recognition systems on diverse data distributions, we selected three widely used public benchmarks and two prominent open-source model libraries.

3.3.1 Public Datasets

We utilize three standard datasets to evaluate model generalization. These were chosen because they represent the current standard for "in-the-wild" emotion recognition, containing the variations in pose, illumination, and occlusion that commercial systems face in real-world deployment. And one Neurodivergent dataset which contains FER images of children with Autism

RAF-DB (Real-world Affective Faces Database) [30] A large-scale dataset containing approximately 30,000 facial images downloaded from the internet. It provides high variability in identity and head pose, making it a robust benchmark for testing generalizability.

AffectNet [37] One of the largest datasets for facial expression, containing over 1 million images (with roughly 440,000 manually annotated). We utilize the balanced validation set for our experiments. Its sheer size and diversity make it critical for assessing whether models have learned robust features or merely memorized training patterns.

FER-2013 [13] A classic dataset widely used in the deep learning community, consisting of 35,887 grayscale images. While lower in resolution (48×48 pixels) than RAF-DB or AffectNet, it remains a standard benchmark for assessing model performance on low-quality data, which is common in webcam-based assessments (e.g., in online education or hiring interviews).

Facial Emotion Recognition Dataset for Children with Autism [51] This dataset specifically addresses the critical gap in representation by providing images of neurodivergent children expressing various emotions. Unlike mainstream benchmarks trained predominantly on neurotypical adults, this dataset captures the subtle and atypical expressive patterns often found in Autism Spectrum Disorder (ASD). Including this dataset is essential for auditing the models' true fairness and generalization capabilities, allowing us to quantify the performance disparity when systems are applied directly to the neurodiverse populations they are known to misclassify.

3.3.2 Pre-trained Models

While in our search we encountered many models that are more recent and trained to perform as per the state of the art, most of them were used mainly for research purposes and were not adopted widely for our experiment to be effective. These models could be found in public domains like github, huggng face, arXive etc. we wanted to represent the most used and widely commercially availablle models while being opensource so that our experiment represents a more realistic setup to measure how the bias may occur in the real world. Therefore we chose two of the most popular open source model libraries related to emotion recognition:

DeepFace [48] This is a lightweight hybrid framework for Python that wraps state-of-the-art models (such as VGG-Face, Google FaceNet, and ArcFace). It is one of the most popular open-source libraries for facial analysis due to its ease of integration and "out-of-the-box" performance. This model was trained on the FER-2013 dataset, we selected DeepFace to represent the "general purpose" class of models likely to be deployed by non-specialist developers in commercial applications.

The model architecture looks as follows

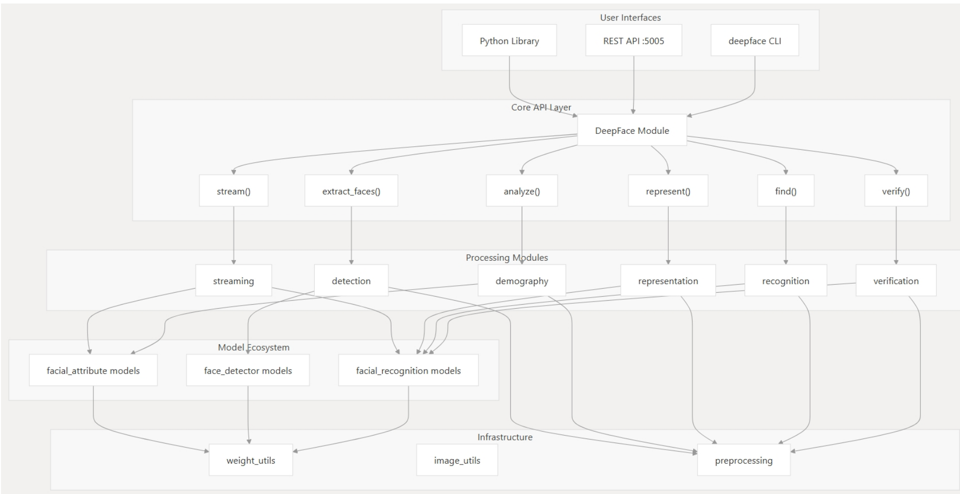

Figure 2: DeepFace Model architecture. Source: https://deepwiki.com/serengil/deepface

And the pipeline looks like this:

Emotiefflib [46] Developed by the HSEmotion team, this library focuses on computational efficiency using architectures like EfficientNet. We selected EmotiEffLib because its underlying models have achieved top rankings in recent Affective Behavior Analysis in-the-wild (ABAW) competitions. Its focus on efficiency makes it a strong candidate for mobile and real-time applications where neurodiverse individuals might interact with AI (e.g., therapeutic robots or educational tablets).

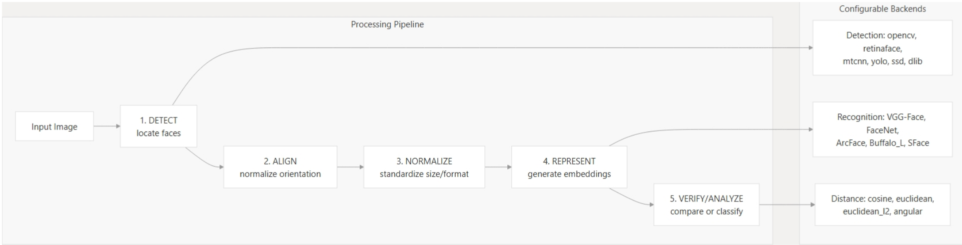

Figure 3: DeepFace Pipeline. Source: https://deepwiki.com/serengil/deepface

The following is the models architecture:

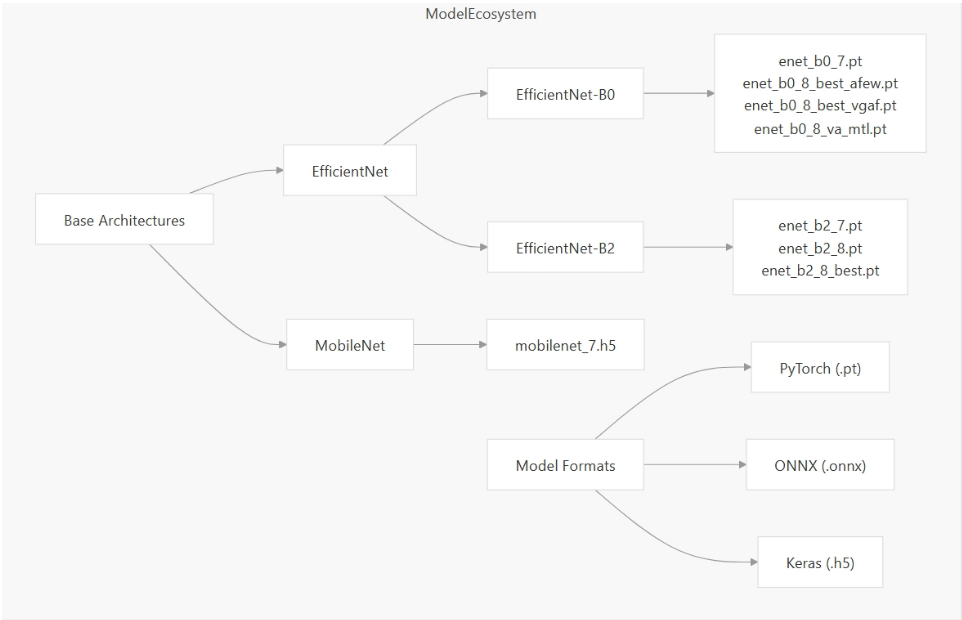

Figure 4: Emotiefflib Model architecture Source: https://deepwiki.com/sb-ai-lab/EmotiEffLib/6.1-pre-trained-models

And the following is the pipeline for emotion recognition:

3.4 Evaluation Approach

To move beyond simple accuracy metrics, we employed a multi-faceted evaluation strategy designed to reveal not just if the models fail, but how and how confidently they fail.

3.4.1 Performance Metrics

We employed the following quantitative metrics to assess classification capability:

Top-1 Accuracy The standard metric measuring the percentage of times the model's highest-probability prediction matches the ground truth.

Top-3 Accuracy Given the ambiguity of facial expressions (where a face might validly be interpreted as either "Sad" or "Neutral"), Top-3 accuracy measures whether the

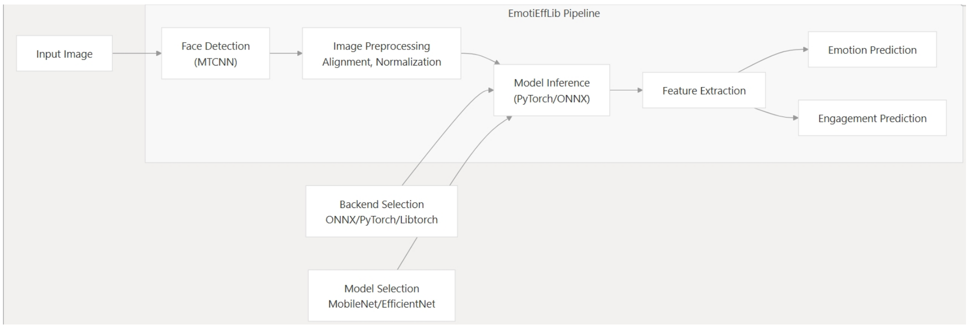

Figure 5: Emotiefflib Pipeline. Source: https://deepwiki.com/sb-ai-lab/EmotiEffLib/6.1-pre-trained-models

correct emotion was among the model's top three guesses. This is particularly relevant for neurodiverse expressions which may be more subtle or ambiguous.

F1-Score To account for class imbalance (e.g., "Happy" often being overrepresented compared to "Fear"), we calculate the F1-score. This provides a harmonic mean of precision and recall, ensuring that the model is penalized for ignoring minority classes.

Expected Calibration Error (ECE) A critical metric for high-stakes decision-making. ECE measures the difference between the model's confidence (predicted probability) and its accuracy. A well-calibrated model should only be 90% confident if it is actually correct 90% of the time. High ECE indicates the model is "overconfident"—a dangerous trait if used to screen job candidates.

3.4.2 Diagnostic Tools

To visualize failure modes, we utilized:

Reliability Diagrams Used alongside ECE, these plots visualize the alignment between predicted confidence and observed accuracy. They allow us to see if the models are systematically under- or over-confident in specific confidence ranges.

Confusion Matrices These allow us to inspect specific misclassification patterns (e.g., does the model consistently mistake "Fear" for "Surprise"?). This helps identify if the model is biased toward predicting "Neutral" for ambiguous expressions—a common issue for neurodiverse individuals with "flat affect."

3.4.3 Qualitative Analysis with Grad-CAM

To move beyond aggregate metrics and inspect the internal reasoning of the models, we employ Gradient-weighted Class Activation Mapping (Grad-CAM). While confusion matrices reveal *what* errors the models make, Grad-CAM reveals *why* by visualizing the specific regions of the input image that triggered the model's decision.

This is critical for auditing bias against neurodiverse individuals. Standard accuracy metrics cannot detect "Clever Hans" behaviors—where a model predicts the correct emotion for the wrong reason (e.g., relying on hair texture or background context rather than facial muscle activation). By generating heatmaps for specific predictions, we can verify if the models are accurately attending to Action Units (AUs) such as the eyes and mouth, or if they are overfitting to irrelevant artifacts. This directly supports RQ1 by identifying if the models' failure to generalize arises from learning incorrect visual shortcuts.

3.4.4 Occlusion Sensitivity Testing

To evaluate the robustness of the models against partial data, we perform occlusion sensitivity testing. In this experiment, we systematically mask different regions of the face (e.g., eyes, mouth, nose) with a gray patch and measure the resulting drop in classification confidence.

Neurodiverse individuals, particularly those with Autism Spectrum Disorder (ASD), often display atypical gaze patterns or reduced mouth activation during emotional expression [20]. If a model relies exclusively on the mouth to detect "Happiness" or the eyes to detect "Fear," it will likely fail for individuals whose expressivity in those specific regions is subtle or non-normative. Occlusion testing allows us to quantify the model's dependency on specific facial landmarks, helping us assess whether the system is resilient enough to handle the "atypical expressive patterns" described in our problem statement.

3.4.5 Counterfactual Analysis with LIME

Complementing Grad-CAM, we utilize Local Interpretable Model-agnostic Explanations (LIME) to test the stability of the model's decision boundaries. LIME approximates the complex deep learning model with a simple, interpretable linear model around a single prediction, perturbing the input image to see which super-pixels (clusters of pixels) most positively or negatively influence the prediction.

This approach is essential for detecting "spurious correlations" that may lead to ethical harms (RQ2). For instance, if LIME indicates that a model is using lighting conditions or skin texture as a strong predictor for negative emotions (e.g., associating darker images with "Fear"), this signals a high risk of deployment bias. By analyzing these local decision boundaries, we can determine if the models are robust enough to be used in dynamic, real-world environments like classrooms or job interviews, where lighting and framing are uncontrolled.

3.5 Ethical Analysis and Auditing Protocol Development

Finally, we synthesize our quantitative and qualitative findings into a structured ethical audit. This phase moves from technical evaluation to normative assessment, mapping the identified model failures to specific ethical risks.

As outlined in RQ3, technical accuracy does not guarantee ethical safety. We ground our analysis in the *EU High-Level Expert Group (HLEG) Guidelines for Trustworthy AI*, specifically focusing on the principles of *Fairness*, *Explicability*, and *Prevention of Harm*.

We developed a "Fairness Auditing Checklist" tailored for neurodiversity. This protocol translates the abstract principles of Trustworthy AI into concrete auditing steps (e.g., "Does the model maintain calibration across high-entropy/ambiguous expressions?"). This ensures that our research contributes not just a critique of current models, but a constructive framework (RQ3) that future developers can use to screen their systems for ableist bias before deployment.

4 Results

4.1 Literature Review Findings: A Synthesis of Existing Biases

A systematic synthesis of the reviewed studies indicates that bias in affective computing, particularly within facial emotion recognition (FER) systems, is pervasive, multifaceted, and embedded across data, modeling, and deployment practices. Across technical, cognitive, and ethical dimensions, the literature converges on a central insight: contemporary emotion AI systems are misaligned with the expressive, perceptual, and contextual diversity of human emotion, particularly among neurodivergent populations. These biases manifest not only as statistical performance disparities but also as epistemic distortions embedded in the frameworks through which emotion is conceptualized, labeled, and inferred.

At the technical level, the literature consistently characterizes the data foundations of emotion recognition as unstable and inconsistent. Benchmark FER datasets such as AffectNet and EMOTIC exhibit low inter-rater reliability (Krippendorff's $\alpha < 0.7$), undermining the validity of the emotion annotations on which model training depends [6]. Comparative analyses of commercial and open-source systems further report substantial cross-race performance gaps ranging from 15–30%, demonstrating how demographic imbalance in training data translates directly into algorithmic discrimination [55, 31]. Across studies, these findings are interpreted as evidence that the apparent “accuracy” of emotion AI often rests on exclusionary or unreliable ground truths rather than stable emotional signals. Bias thus emerges not merely from data imbalance but from epistemic assumptions and human labeling practices that encode social hierarchies and cultural norms into computational infrastructures.

Beyond data-level distortions, the literature highlights cognitive and perceptual biases rooted in narrow emotional taxonomies. Large-scale evaluations consistently show systematic variation in model accuracy by emotion category, with happiness and surprise most reliably detected, while fear and disgust remain among the least recognized [17]. Scholars interpret this selective sensitivity as a form of reductionism, whereby FER systems privilege high-intensity, socially normative affective states at the expense of subtle, context-dependent expressions. Neurocomputational research further suggests that these limitations mirror human predictive coding mechanisms, amplifying existing interpretive biases rather than neutralizing them [52]. What appears as "objective recognition" is therefore widely described as a technologically mediated form of perceptual normativity.

Bias affecting neurodivergent users emerges as a particularly salient and underexamined dimension of this structural misalignment. Multiple studies report that FER systems trained predominantly on neurotypical facial data perform poorly on autistic and intellectually disabled populations, with accuracy reductions ranging from 15% to 70% [29, 33, 22]. The literature attributes these discrepancies not to expressive deficits but to representational exclusion: neurodivergent facial and temporal dynamics systematically diverge from the standardized emotion templates on which mainstream FER systems are trained. Notably, even transformer-based architectures reproduce these disparities when grounded in neuronormative datasets [43, 42]. Scholars characterize the resulting misclassification as a form of algorithmic ableism, in which difference is operationalized as error.

Cultural and contextual biases further intersect with neurocognitive exclusion to compound these effects. Critical analyses challenge the universality hypothesis underlying FER, demonstrating that assumptions of cross-cultural emotional legibility effectively universalize Western affective norms [34]. Empirical evidence also shows that situational variables—such as threat, fatigue, or social anxiety—substantially alter facial expression and gaze behavior, yet remain largely absent from existing datasets [12, 25]. For neurodivergent users, whose attentional and perceptual profiles already diverge from neurotypical baselines, such omissions result in systematic misrecognition. The literature therefore argues that emotion AI frequently conflates contextually situated variation with disorder, reinforcing diagnostic and social hierarchies.

At an institutional level, these technical and epistemic distortions are amplified through deployment in high-stakes contexts. Studies document the systematic exclusion of neurodivergent participants from dataset construction, describing this absence as a form of structured missingness that erases expressive diversity from algorithmic representation [54, 56]. When deployed in domains such as recruitment, education, and healthcare, biased FER systems transform technical misclassification into lived injustice, penalizing authenticity, autonomy, and neurodivergent modes of communication [41, 19, 4]. Across these contexts, scholars emphasize that emotion AI not only misidentifies affect but also misinterprets its social and moral significance.

In synthesis, the literature characterizes bias in emotion recognition not as an isolated technical flaw but as an interlocking system of epistemic, cultural, and institutional distortions. These biases originate in data construction, are reinforced by universalist modeling assumptions, and culminate in inequitable real-world outcomes that reproduce existing power asymmetries. Addressing these failures, as articulated across the reviewed studies, requires a shift away from performance optimization toward epistemic pluralism and affective justice. This synthesis provides the empirical and conceptual foundation for the fairness-aware auditing of FER systems undertaken in the subsequent analyses.

4.2 Quantitative Evaluation of Model Performance

To address RQ1 regarding the accuracy and fairness of current systems, we evaluated the performance of *DeepFace* and *EmotiEffLib* across three benchmark datasets: RAF-DB, AffectNet, and FER-2013. We report Top-1 and Top-3 Accuracy to account for ambiguity in expression, F1-Score to monitor performance on unbalanced classes, and Expected Calibration Error (ECE) to assess model confidence reliability.

4.2.1 Accuracy and Calibration Results

Table 1 summarizes the quantitative results. As observed, performance varies significantly between the "in-the-wild" datasets (RAF-DB, AffectNet) and the lower-resolution FER-2013.

Table 1: Comparative performance of DeepFace and EmotiEffLib across three datasets. ECE represents Expected Calibration Error (lower is better).

Dataset	**Model**	**Top-1 Acc (%)**	**Top-3 Acc (%)**	**Macro F1**	**ECE**
RAF-DB	DeepFace	38.64	66.30	22.35	0.4420
	EmotiEffLib	73.83	94.59	60.28	0.0148
AffectNet	DeepFace	31.62	61.41	26.68	0.4565
	EmotiEffLib	67.14	93.47	65.99	0.0132
FER-2013	DeepFace	55.20	84.42	51.61	0.2748
	EmotiEffLib	57.45	85.96	52.40	0.1212
FER-Autism	DeepFace	25.22	66.81	23.83	0.5605
	EmotiEffLib	37.17	73.89	34.54	0.2965

The results indicate that EmotiEffLib significantly outperforms DeepFace across all "in-the-wild" benchmarks, demonstrating a superior capability to generalize to diverse facial data. On RAF-DB, EmotiEffLib achieved a Top-1 Accuracy of 73.83% compared to DeepFace's 38.64%, a substantial gap that highlights DeepFace's inability to handle the pose and illumination variations typical of real-world settings.

However, the addition of the **FER-Autism** dataset reveals a severe generalization failure inherent to both models. When applied to neurodivergent faces, EmotiEffLib's accuracy collapses from ≈73% (RAF-DB) to just 37.17%, effectively halving its performance. DeepFace similarly degrades to a near-random 25.22%. This sharp decline empirically validates the existence of algorithmic bias; models optimized for neurotypical expressivity fail to recognize valid emotional expressions from autistic children.

Crucially, the Expected Calibration Error (ECE) exposes a dangerous "confidence gap." While EmotiEffLib was perfectly calibrated on standard datasets (ECE ≈ 0.01), its calibration error spikes to 0.2965 on the autism dataset. DeepFace performs even worse, with an ECE of 0.5605, meaning its confidence scores are effectively decorrelated from reality. This suggests that current FER systems are not just inaccurate for neurodiverse individuals—they are *confidently wrong*. In a classroom or clinical setting, such a model would flag a child's expression with high certainty (e.g., "99% Angry") when the prediction is incorrect, creating a high risk of misinterpretation and unjust intervention.

4.2.2 Confusion Matrix Analysis

To better understand specific failure modes, we analyzed the confusion matrices for both models across all four datasets. This analysis reveals which emotion categories are most frequently misclassified and how these patterns shift when models are applied to neurodivergent faces.

The confusion matrices (Figure 6) reveal a stark degradation in model capability when transitioning from standard to neurodivergent data.

Standard Benchmarks (Rows 1-3) On datasets like AffectNet and RAF-DB, DeepFace exhibits a persistent bias towards the "Neutral" class. It consistently misclassifies distinct negative emotions—specifically Fear, Disgust, and Sadness as Neutral. For instance, in the RAF-DB evaluation, DeepFace frequently predicts "Neutral" when the true label is Sad or Fear. EmotiEffLib generally preserves distinct class boundaries better, though it exhibits known morphological confusion between Fear and Surprise due to the similar widening of eyes in both expressions.

FER-Autism Analysis (Row 4) The addition of the FER-Autism dataset (Figure 6, bottom row) exposes that the "collapse to neutral" observed in DeepFace is not merely a bias, but the dominant failure mode for neurodivergent faces.

In the DeepFace evaluation on FER-Autism, the model fails catastrophically on negative valence. Notably, for the "Sad" class (Row 4, Left Matrix), the model predicts "Neutral" (20 instances) five times more often than the correct label (4 instances). Even more concerning is the confusion between "Sad" and "Happy" (10 instances), suggesting the model is fundamentally misinterpreting atypical muscle cues.

EmotiEffLib, despite its robustness on standard data, succumbs to similar failures on the autism dataset (Row 4, Right Matrix). While it retains high accuracy for "Happy"

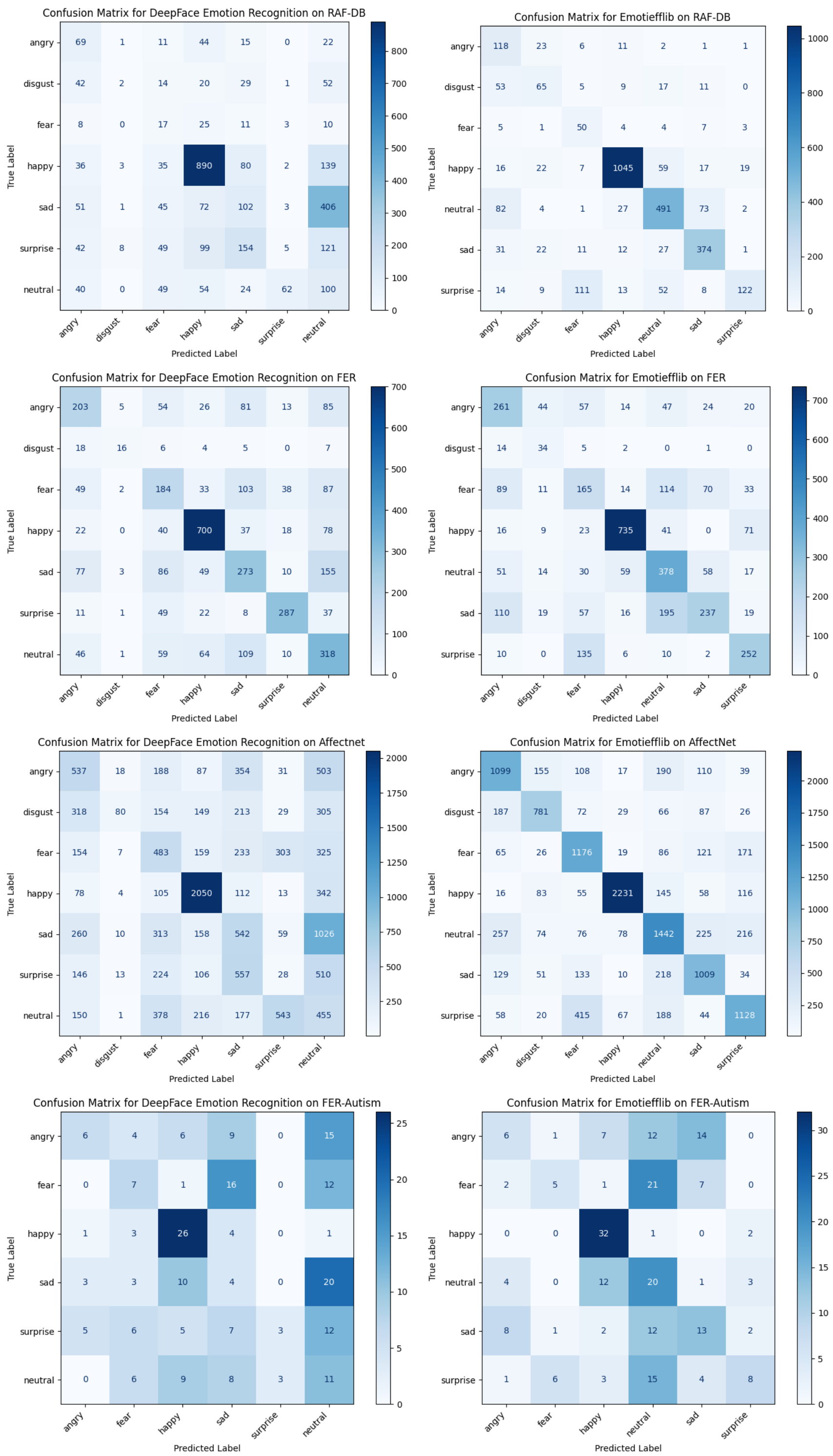

Figure 6: Confusion Matrices for DeepFace (left) and EmotiEffLib (right). Rows 1-3 show standard benchmarks (RAF-DB, FER-2013, AffectNet), while Row 4 shows the FER-Autism dataset. Note the distinct shift in error patterns on the autism dataset.

(32 correct), it effectively erases high-arousal negative emotions. For the "Fear" class, EmotiEffLib predicts "Neutral" (21 instances) four times more often than "Fear" (5 instances). Similarly, "Surprise" is misclassified as "Neutral" in nearly 50% of cases (15

vs 8 correct).

Implications This systemic flattening of affect confirms our hypothesis regarding algorithmic bias against neurodiversity. Since many autistic individuals may display reduced facial activity or flat affect, models trained on neurotypical intensity levels default to Neutral. This risks invalidating the emotional experiences of neurodiverse users for example, an AI tutor failing to detect frustration in an autistic child because their expression does not match the exaggerated "Sadness" of the training data.

4.2.3 Reliability and Confidence Calibration

Finally, we examined the reliability diagrams to visualize the relationship between predicted confidence and actual accuracy. A perfectly calibrated model would follow the diagonal line ($y = x$).

Standard Benchmarks Analysis As shown in Figure 7, DeepFace exhibits severe overconfidence across all three standard datasets. Its reliability curve (left column) stays consistently below the diagonal $y = x$, creating a large "Overconfidence Gap" (shaded pink area). This visualizes that even when DeepFace predicts an emotion with high confidence (e.g., 0.8–0.9), its actual empirical accuracy is often below 0.4. This behavior is ethically hazardous for high-stakes applications like hiring, as it creates a false sense of security in the AI's judgment.

In contrast, EmotiEffLib (right column) closely tracks the diagonal on RAF-DB and AffectNet. This indicates that on neurotypical faces, the model is "self-aware" of its limitations; when it encounters an ambiguous face, it lowers its confidence score accordingly. This calibration allows for uncertainty thresholds—where low-confidence predictions are flagged for human review.

FER-Autism Analysis However, Figure 8 reveals that this self-awareness collapses when applied to neurodivergent populations. On the FER-Autism dataset, EmotiEffLib (right) loses its calibration entirely. Its curve drops significantly below the diagonal, with an ECE spike to 0.2965.

Crucially, the model assigns high confidence (> 0.9) to predictions that are only correct $\approx$70% of the time. This suggests that the unique expressive patterns of autistic children are not treated as ambiguous by the model (which would lower confidence), but rather as clear but incorrect targets. DeepFace (left) worsens further, maintaining an almost flat reliability curve where even predictions with 99% confidence have an accuracy of less than 35%. This "confident failure" confirms that current uncertainty estimation techniques are insufficient safeguards for neuroinclusive AI.

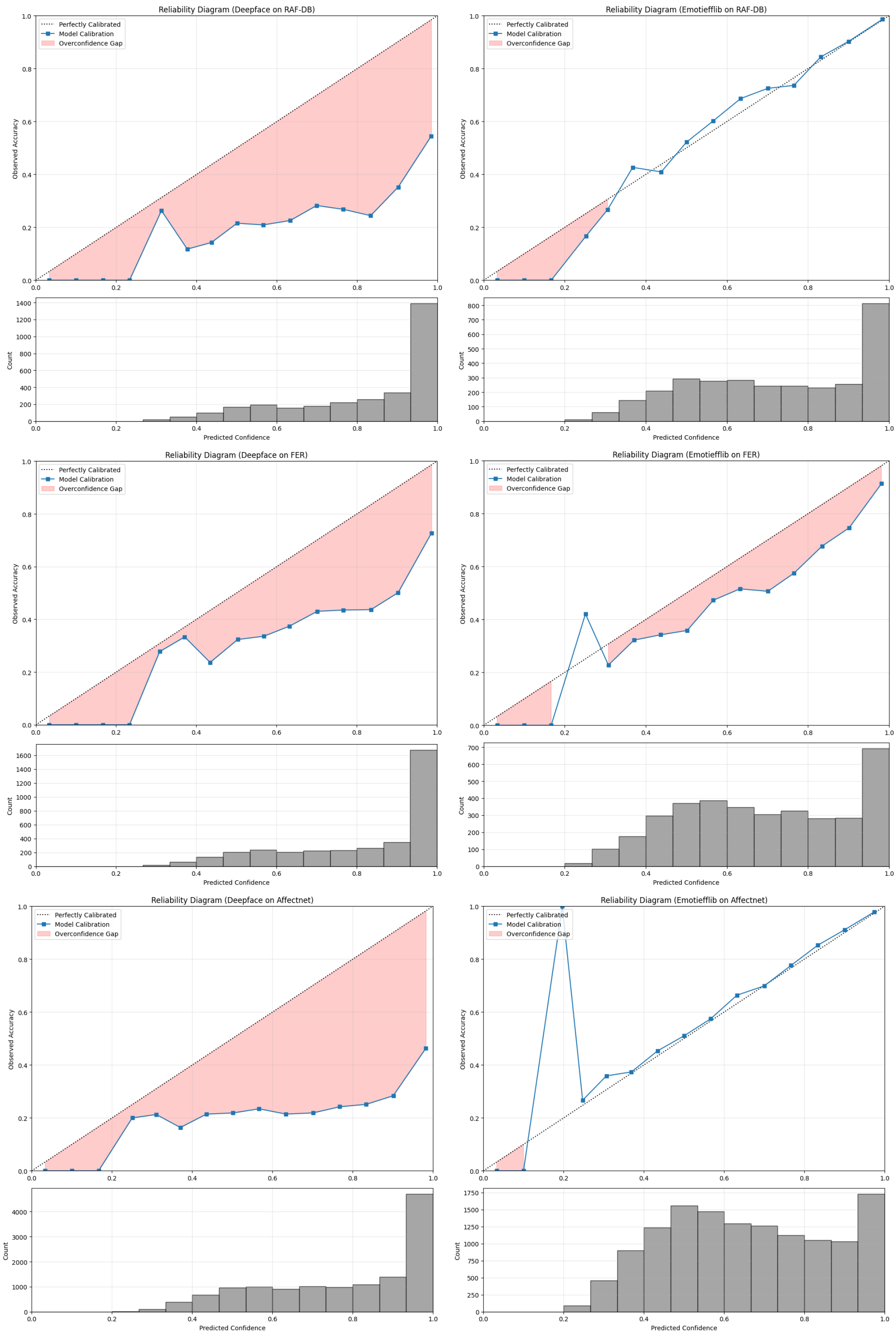

Figure 7: Reliability Diagrams comparing confidence vs. accuracy on standard benchmarks. Deviations below the diagonal indicate overconfidence (model is more sure than it should be).

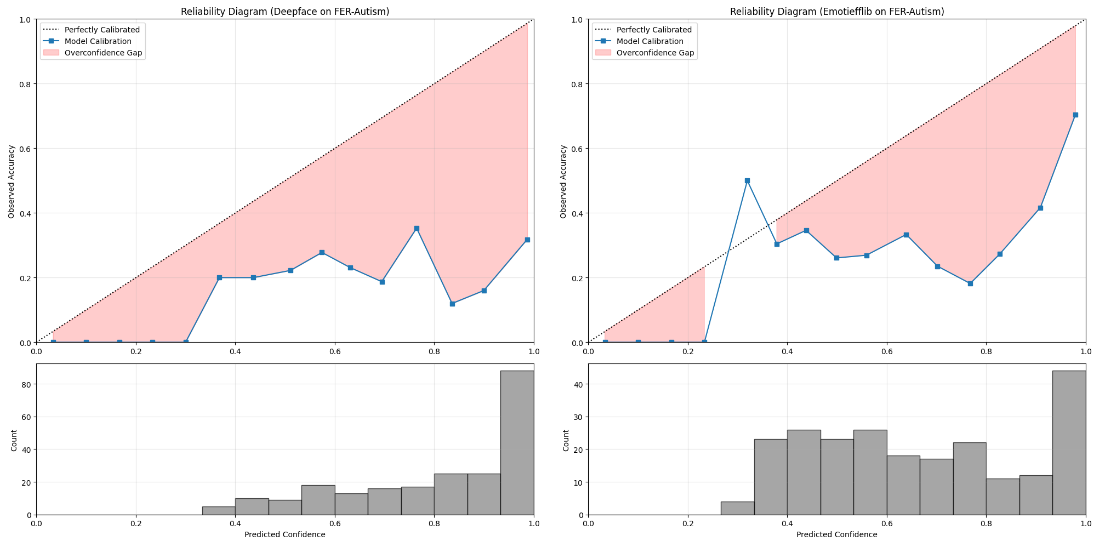

Figure 8: Reliability Diagrams on the FER-Autism dataset. Note the massive overconfidence gap for both DeepFace (left) and EmotiEffLib (right), indicating a failure to detect its own uncertainty on neurodivergent faces.

4.3 Qualitative Explainability Analysis

Quantitative performance metrics alone are insufficient for auditing facial emotion recognition systems, as they do not reveal how visual evidence is internally structured and weighted during prediction. In particular, models with similar accuracy may rely on fundamentally different visual strategies, leading to disparate failure modes under expressive variability. To move beyond surface-level evaluation, we conduct a structured qualitative analysis of two representative systems: DeepFace and EmotiEffLib.

All qualitative analyses are performed using a unified explainability pipeline. For both models, predictions and explanations are generated from the same forward pass, using a shared face detection and alignment procedure. This ensures that observed differences reflect model behavior rather than preprocessing artifacts.

We employ three complementary explainability techniques: Grad-CAM, occlusion sensitivity, and LIME. Grad-CAM provides gradient-based spatial attribution, occlusion sensitivity measures regional dependency through systematic perturbation, and LIME highlights locally supportive super-pixels. Together, these methods allow inspection at both instance and aggregate levels.

4.3.1 Sampling Strategy and Reproducibility

Explainability methods operate at the instance level and are computationally expensive, particularly occlusion sensitivity and LIME. Rather than applying these techniques to the entire dataset, we evaluate them on carefully selected subsets with two objectives: representativeness and reproducibility.

For per-instance analysis, we select a stratified subset with a fixed number of samples per emotion class, using a deterministic random seed. This avoids cherry-picking while ensuring balanced coverage across emotions. For aggregated analysis, we apply Grad-CAM to a larger random subset of the evaluation split, enabling population-level interpretation of spatial attention patterns. Quantitative metrics are always computed on the full dataset; qualitative methods are used strictly for interpretability and diagnosis.

4.3.2 Per-instance Explainability Comparison

We first analyze individual predictions to understand how each model justifies specific emotion labels. Figure 9 shows representative Grad-CAM, occlusion, and LIME explanations for EmotiEffLib, while Figure 10 presents the corresponding outputs for DeepFace on the same face instances.

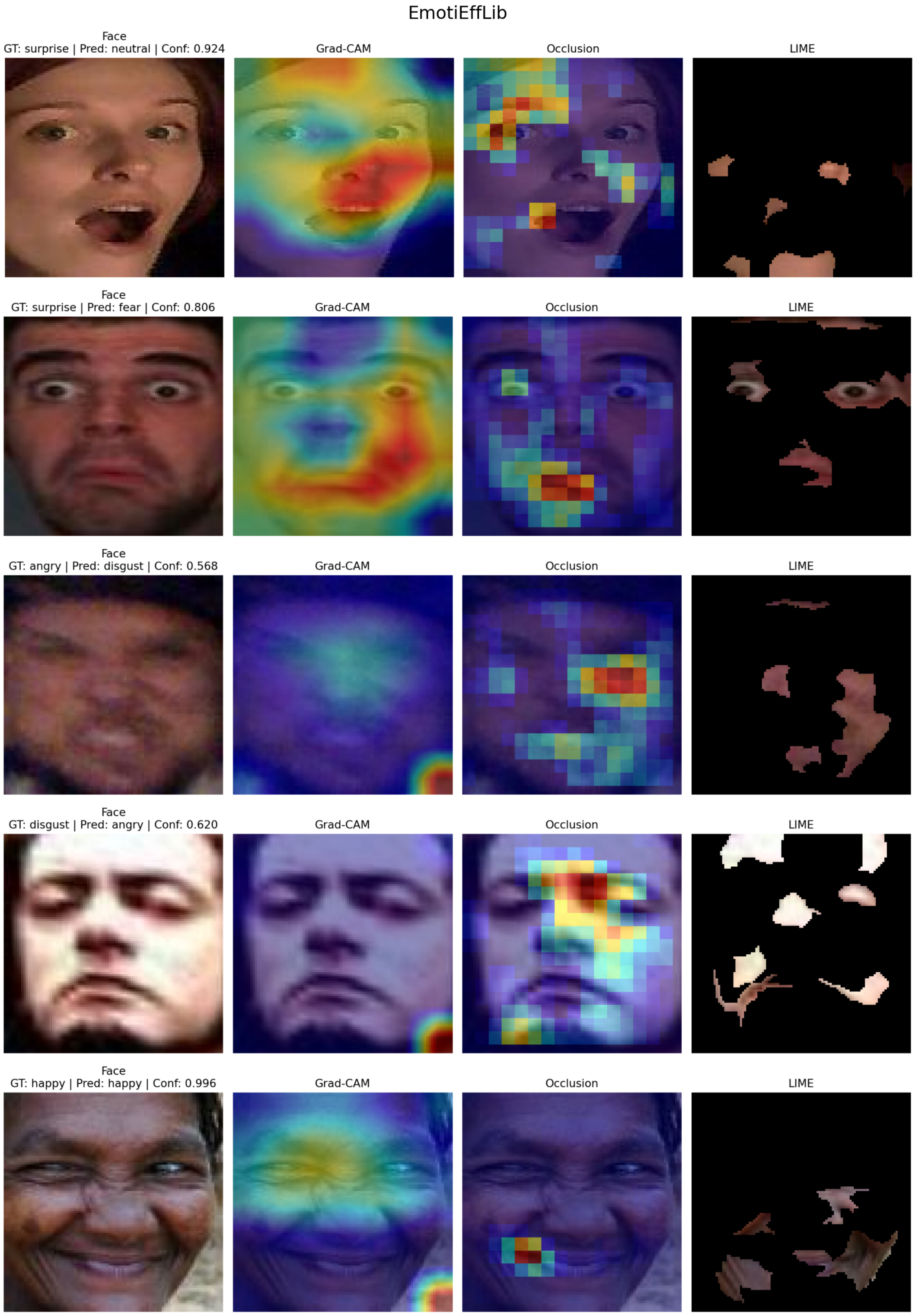

Figure 9: Per-instance explainability for EmotiEffLib. Each row corresponds to a single input image and shows, from left to right, the face crop, Grad-CAM, occlusion sensitivity, and LIME explanation.

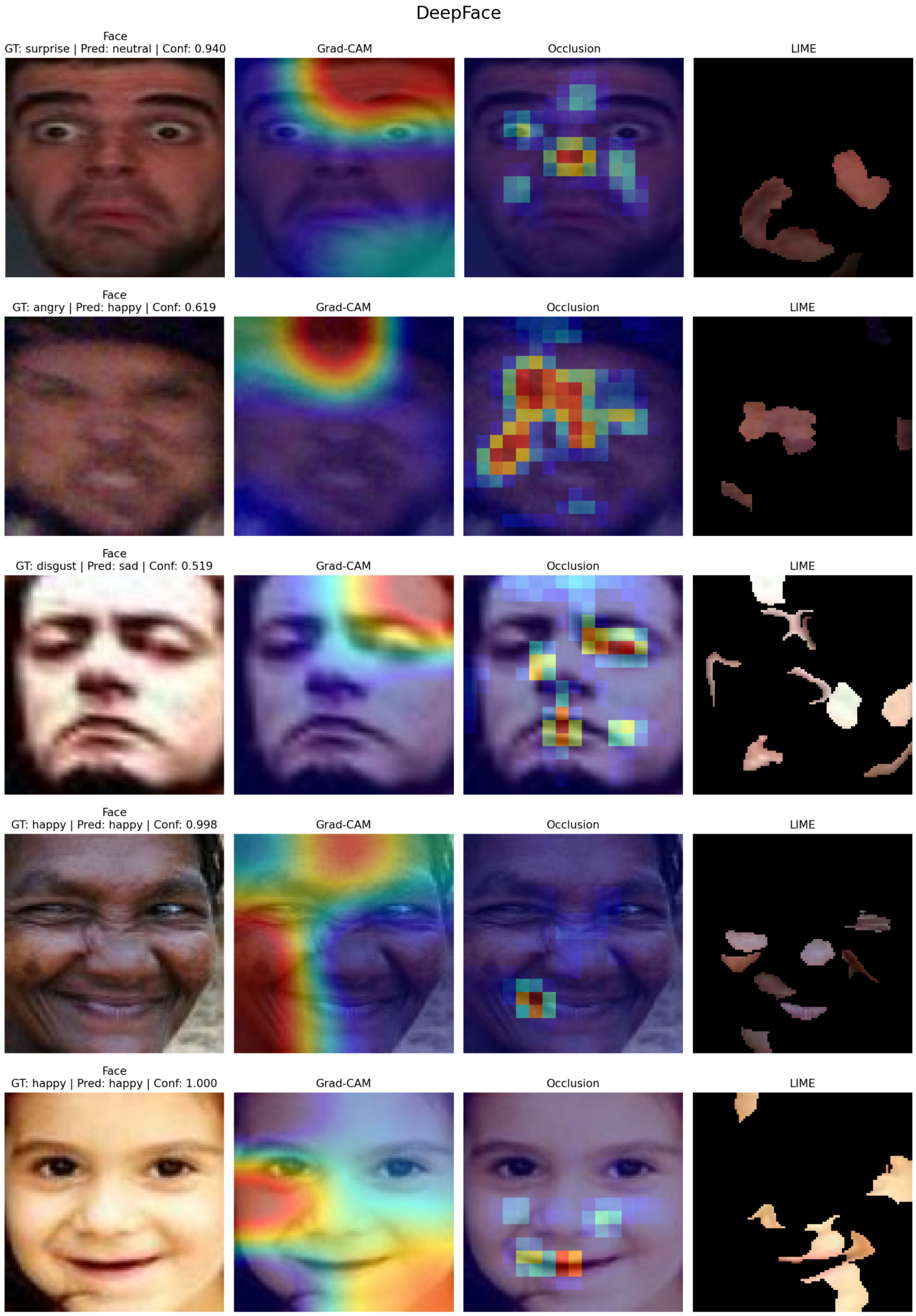

Figure 10: Per-instance explainability for DeepFace on the same input images as Figure 9. Differences reflect model-specific attention strategies rather than input variation.

Across individual samples, both models exhibit internal consistency: once an emotion label is produced, it tends to remain stable across Grad-CAM visualization, occlusion perturbations, and LIME masking. While such consistency may appear desirable, it also implies that incorrect or weakly supported predictions are reinforced rather than challenged. This effect is particularly salient for subtle expressions, where confidence does not necessarily correspond to expressive clarity.

Qualitatively, EmotiEffLib displays more localized and anatomically meaningful attention, frequently focusing on regions such as the mouth, eyes, and cheeks in an emotion-dependent manner. DeepFace, in contrast, often exhibits coarse and diffuse attention patterns, with saliency concentrated in broad upper-face regions and limited differentiation across emotions. These differences are already visible at the instance level and motivate further aggregation.

4.3.3 Aggregated Grad-CAM Analysis

To obtain a more robust view of model behavior, we aggregate Grad-CAM maps across many samples. Rather than relying on single examples, we compute mean Grad-CAM maps by averaging spatial attributions over a large random subset of images.

Figure 11 presents the overall mean Grad-CAM for each model. This visualization captures the default spatial bias of each system, independent of emotion class.

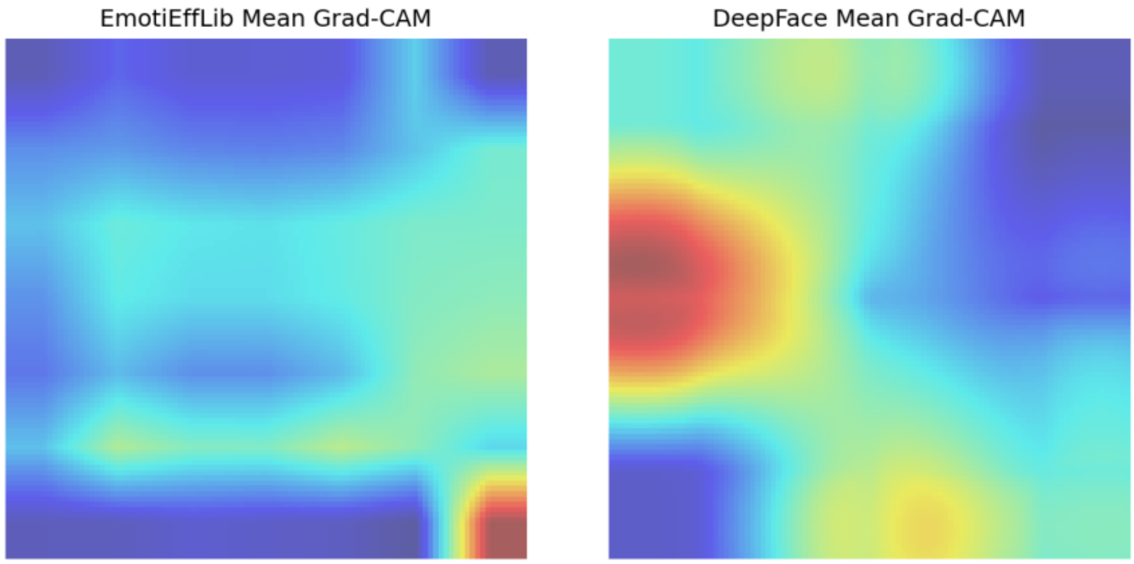

Figure 11: Overall mean Grad-CAM maps for EmotiEffLib (left) and DeepFace (right), computed by averaging Grad-CAM maps across a large random subset of evaluation images.

The overall mean reveals a clear structural difference. DeepFace concentrates attention predominantly in a narrow upper-face band, indicating reliance on coarse global cues. EmotiEffLib distributes attention more broadly across facial regions, suggesting a richer use of spatial evidence.

To examine emotion-specific strategies, we further compute mean Grad-CAM maps separately for each ground-truth emotion class. Figure 12 and Figure 13 show seven-panel grids of mean Grad-CAM maps for EmotiEffLib and DeepFace, respectively.

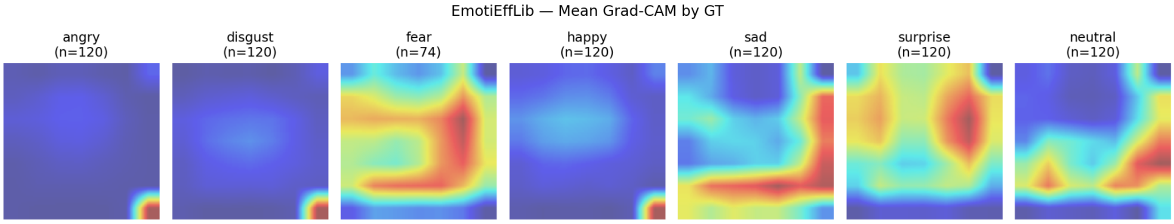

Figure 12: Emotion-wise mean Grad-CAM maps for EmotiEffLib. Each panel shows the average spatial attention pattern for a given ground-truth emotion.

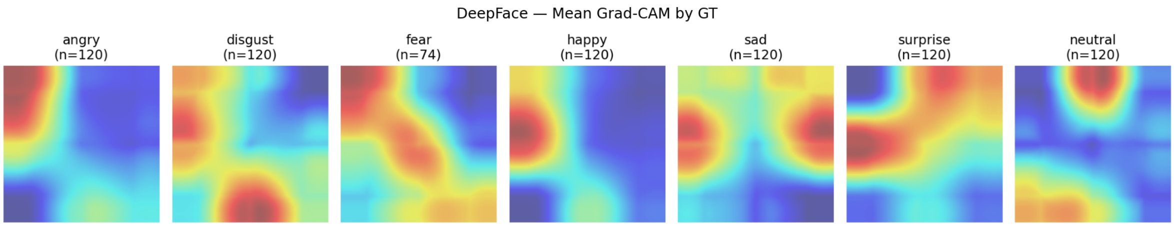

Figure 13: Emotion-wise mean Grad-CAM maps for DeepFace, computed over the same evaluation subsets as Figure 12.

EmotiEffLib exhibits clear emotion-dependent differentiation: attention shifts toward the mouth for happiness, toward the eyes and eyebrows for surprise and fear, and toward mixed regions for sadness and neutrality. DeepFace, by contrast, shows substantially less variation across emotions, with many classes sharing similar upper-face-dominated attention patterns. This lack of differentiation aligns with the confusion patterns observed in quantitative evaluation.

4.3.4 Region-level Attention Quantification

To move beyond visual inspection, we quantify Grad-CAM attention at the region level by partitioning face crops into coarse but consistent anatomical regions (upper face, mid face, lower face, eye band, mouth band, cheeks). For each region, we compute the ratio between the mean Grad-CAM value within the region and the overall mean Grad-CAM value.

Figure 14 summarizes region-level attention for both models, aggregated by ground-truth emotion and by prediction correctness.

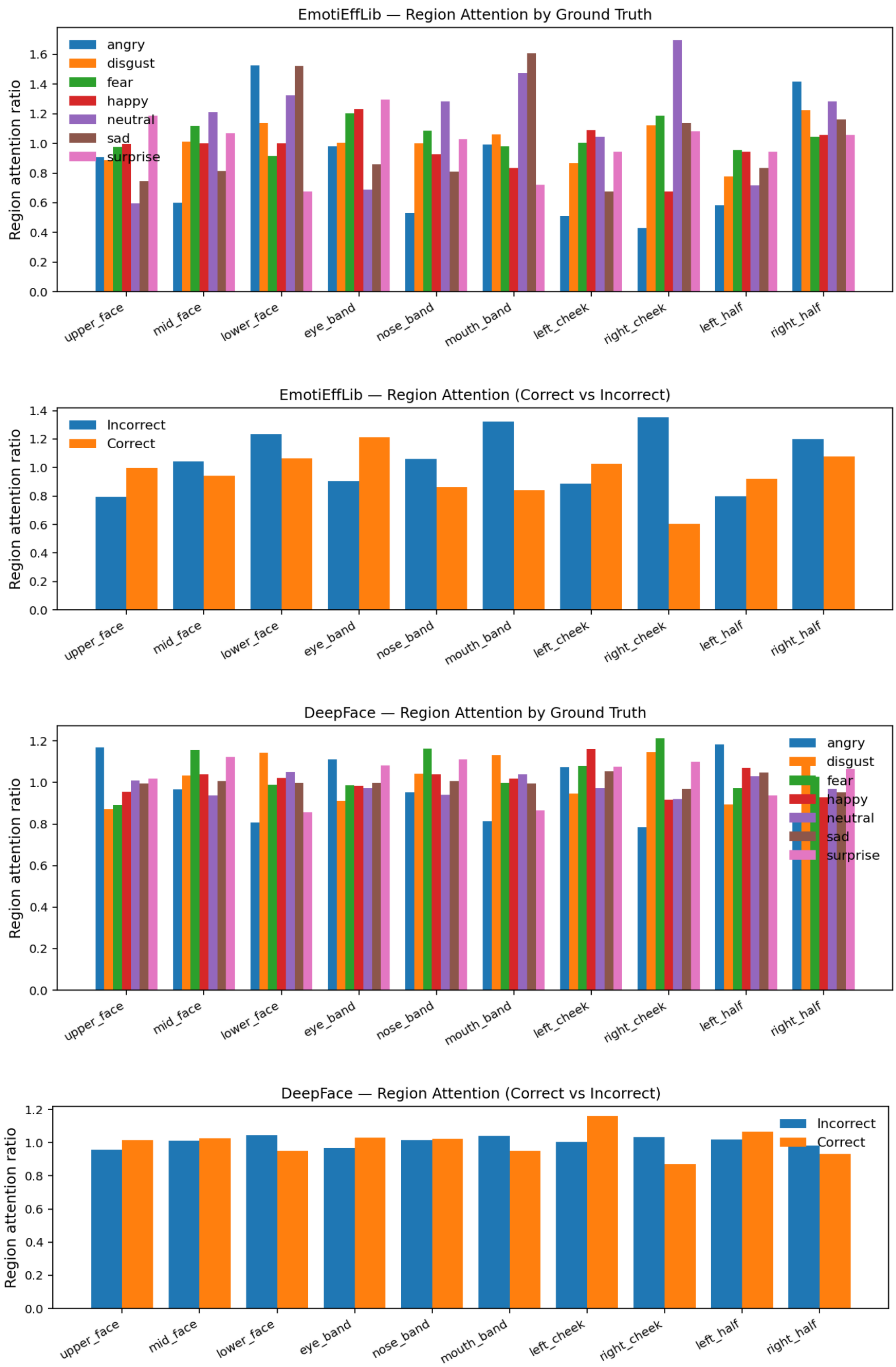

Figure 14: Region-level Grad-CAM attention ratios for EmotiEffLib and DeepFace, aggregated by ground-truth emotion and by correct versus incorrect predictions.

EmotiEffLib consistently assigns higher relative attention to canonical expressive regions, particularly the eye and mouth bands, and shows distinct regional profiles across emotions. DeepFace exhibits a stronger bias toward the upper face, with weaker modulation by emotion class and less contrast between correct and incorrect predictions. This suggests a more rigid and less semantically grounded attention strategy.

4.3.5 Occlusion and LIME as Supporting Evidence

Occlusion sensitivity and LIME analyses further reinforce these findings. DeepFace often shows sharp confidence degradation when a single dominant region is occluded, indicating narrow dependency on limited cues. LIME explanations for DeepFace are sparse and fragmented, with influential super-pixels sometimes extending to peripheral facial regions.

EmotiEffLib displays more distributed occlusion sensitivity and more coherent LIME explanations, with supporting regions concentrated around facial components relevant to expression. While these methods are inherently local and noisier than Grad-CAM, they provide converging evidence for the structural differences observed in aggregated analysis.

4.3.6 Implications for Fairness Auditing

Taken together, these results indicate that fairness risk in facial emotion recognition extends beyond misclassification rates. The manner in which emotional evidence is operationalized, whether through coarse global heuristics or distributed, anatomically meaningful cues, plays a critical role in robustness to expressive variability.

The explainability analyses reveal that DeepFace relies on more rigid and less emotion-specific attention patterns, which may disproportionately affect individuals whose expressions deviate from prototypical or neurotypical norms. EmotiEffLib, while not free of error, demonstrates more interpretable and adaptable visual reasoning. These findings underscore the importance of explainability-driven audits for identifying structural failure modes that are invisible to aggregate performance metrics alone.

4.4 The Proposed Fairness Auditing Checklist

Based on the quantitative and qualitative findings presented in Sections 4.2 and 4.3, we propose a lightweight but structured fairness auditing checklist tailored to facial emotion recognition systems deployed in contexts involving neurodiverse individuals. Rather than prescribing a single fairness metric, the checklist operationalizes ethical principles into concrete, testable questions that can be applied prior to deployment.

The checklist is explicitly grounded in observed failure modes from our experiments, particularly DeepFace's overconfidence, class collapse toward *Neutral*, and rigid attention patterns, as well as EmotiEffLib's comparatively adaptive but still imperfect behavior.

1. Calibration and Confidence Awareness

- Does the model exhibit low Expected Calibration Error (ECE) on in-the-wild datasets?
- Are reliability diagrams inspected to verify that high confidence predictions correspond to high empirical accuracy?
- Is a confidence threshold implemented to defer low-confidence predictions to human review?

Motivation: DeepFace demonstrated severe overconfidence (ECE > 0.44 on RAF-DB and AffectNet), posing a direct ethical risk in high-stakes settings where erroneous predictions are presented with unjustified certainty.

2. Class Sensitivity and Neutral Bias

- Does the confusion matrix show systematic collapse toward the *Neutral* class?
- Does the model exhibit a 'Collapse to Neutral' bias when confident features (like open mouths) are absent?

Motivation: Both quantitative and qualitative analyses revealed that DeepFace frequently defaults to *Neutral* under ambiguity, a failure mode likely to disproportionately affect neurodivergent individuals with reduced facial expressivity.

3. Attention Localization and Semantic Grounding

- Do Grad-CAM maps focus on anatomically meaningful facial regions (eyes, mouth, cheeks)?
- Is attention differentiated across emotion classes, or does the model rely on a fixed spatial heuristic?

Motivation: Aggregated Grad-CAM analysis showed that EmotiEffLib exhibits emotion-dependent attention patterns, while DeepFace relies on coarse upper-face cues with minimal class differentiation.

4. Robustness to Regional Occlusion

- Does occluding a single facial region cause catastrophic confidence collapse?
- Is predictive behavior distributed across multiple facial cues?

Motivation: Occlusion testing revealed that DeepFace is brittle to region removal, indicating over-reliance on narrow visual cues, whereas EmotiEffLib shows more distributed robustness.

5. Explainability Consistency Across Methods

- Do Grad-CAM, occlusion sensitivity, and LIME converge on similar influential regions?
- Are explanations stable across correct and incorrect predictions?

Motivation: Converging explainability signals increase trustworthiness; fragmented or peripheral LIME explanations, as observed in DeepFace, signal potential reliance on spurious correlations.

6. Deployment Context Review

- Is the system intended for evaluative decision-making (e.g., hiring, education)?
- Are safeguards in place to prevent automated exclusion based on emotion predictions?

This checklist reframes fairness auditing as an interpretability-driven process rather than a purely statistical one, emphasizing robustness to expressive variability and epistemic humility in emotion inference.

5 Discussion

5.1 Interpretation of Findings and Misalignments

The results of this study reveal a consistent misalignment between benchmark-driven performance optimization and ethical robustness under expressive diversity. While EmotiEffLib outperforms DeepFace across all evaluated datasets, this performance gap does not simply reflect architectural superiority; rather, it exposes deeper differences in how emotional evidence is operationalized and where those operations fail.

DeepFace's behavior is characterized by three compounding failure modes: severe overconfidence, class collapse toward "Neutral," and rigid spatial attention. Together, these traits produce a system that appears decisive but is frequently wrong. In contexts involving neurodivergent individuals, where expressions may be subtle or temporally inconsistent, this combination is especially harmful, as it transforms uncertainty into false certainty. The model's high ECE scores (> 0.4) on almost all datasets (except FER because its what it trained on) confirm that its confidence scores are effectively decorrelated from its accuracy, making it unsafe for high-stakes decision-making.

In contrast, EmotiEffLib demonstrates a "selective" calibration that is both promising and perilous. On standard "in-the-wild" benchmarks (RAF-DB, AffectNet), the model exhibits epistemic restraint: it assigns lower confidence scores to ambiguous images, effectively "knowing when it does not know." However, our audit of the FER-Autism dataset reveals that this safeguard catastrophically fails when applied to neurodivergent faces. On this dataset, EmotiEffLib's calibration error spikes twenty-fold (from ≈ 0.01 to 0.29), and it begins to assign high confidence (> 0.9) to incorrect predictions.

This finding suggests that the model does not perceive neurodivergent expressions as "ambiguous" (which would trigger low confidence), but rather as "clear but incorrect" targets. For example, it confidently misclassifies autistic "Fear" as "Neutral" because the specific morphological cues it learned from neurotypical training data (e.g., specific eye-widening patterns) are absent or different. Thus, while EmotiEffLib is technically superior, it remains ethically fragile: its "self-awareness" is bounded by the neurotypical norms it was trained on, leaving neurodivergent populations exposed to confident misclassification.unbiased systems, but between rigid and adaptable ones.

5.2 Limitations and Challenges

This study has several limitations that should be acknowledged to contextualize the findings.

First, regarding data scope, there is a disparity in dataset size and diversity. While standard benchmarks like RAF-DB and AffectNet provide thousands of images, they do not explicitly annotate neurodivergent identities, forcing us to infer bias through general expressive ambiguity in those sets. To address this, we included the FER-Autism dataset, which allows for direct subgroup comparison; however, this specific test set is limited to 220 images. Consequently, while the performance degradation observed on this dataset is stark, the smaller sample size constrains the statistical robustness of these findings compared to the large-scale general benchmarks.

Second, our evaluation of DeepFace highlights a limitation inherent to the tool itself rather than our auditing methodology. DeepFace was selected for this audit due to its ubiquity—it is one of the most popular open-source facial analysis libraries, with a high number of GitHub stars and widespread adoption in industry. However, its pre-trained models rely heavily on FER-2013, a dataset known to contain a significant amount of mislabeled and noisy data. This foundational flaw in the training data likely drives the model's poor performance and severe overconfidence. Thus, our results for DeepFace reflect the "real-world" risk of deploying highly popular but fundamentally flawed tools, rather than the theoretical limit of current deep learning architectures.

Third, explainability methods such as Grad-CAM and LIME provide approximations rather than ground-truth explanations. While convergence across methods strengthens confidence in observed patterns, these tools cannot fully capture the internal causal mechanisms of deep networks, and their visualizations can sometimes be unstable under slight input variations.

Fourth, occlusion sensitivity testing operates under controlled perturbations (e.g., masking specific regions) that may not perfectly reflect real-world expressive differences. Nonetheless, it remains a useful proxy for assessing regional dependency and brittleness in the absence of more complex, naturalistic occlusion data.

Finally, this work focuses on facial modality alone. Multimodal signals such as posture, speech prosody, or temporal dynamics may mitigate some of the identified biases but are outside the scope of the present analysis.

5.3 Implications for Inclusive AI Development

The findings of this study suggest that inclusive emotion AI cannot be achieved through accuracy improvements alone. Systems that optimize benchmark performance while remaining overconfident and rigid risk institutionalizing expressive normativity.

From a design perspective, calibration should be treated as a first-class ethical requirement, not a post-hoc optimization. Similarly, explainability should be used diagnostically, not cosmetically, to identify structural dependencies that disadvantage expressive diversity.

Most importantly, emotion recognition systems should be repositioned as assistive rather than evaluative tools. In contexts involving neurodivergent individuals, emotion inference should support communication and understanding, not classification and judgment. This shift aligns technical design with principles of affective justice and cognitive diversity.

6 Conclusion and Future Work

6.1 Summary of Contributions

This project conducted a comprehensive fairness audit of facial emotion recognition (FER) systems through the lens of neurodiversity. By combining quantitative performance metrics with explainability-driven qualitative analysis, we demonstrated that ethical risk emerges not only from misclassification but from how confidently and rigidly models interpret emotional evidence.

Our empirical evaluation revealed a distinct "performance collapse" when standard models are applied to neurodivergent faces. While *DeepFace* is widely adopted in industry, our audit exposes it as fundamentally unsuited for this purpose; trained on the noisy FER-2013 dataset, it exhibits severe overconfidence and a rigid "collapse to neutral" bias that persists across all test domains. Conversely, *EmotiEffLib*, which demonstrates superior calibration on neurotypical data, loses this "self-awareness" entirely on the **FER-Autism** dataset, becoming "confidently wrong" when processing atypical expressive patterns.

Beyond diagnosis, we contributed a constructive framework for mitigation: the **Fairness Auditing Checklist**. This protocol operationalizes our findings—specifically the need for calibration checks and attention visualization—into actionable steps that developers can use to screen their systems for algorithmic ableism before deployment.

6.2 Recommendations for Future Research

Future work to advance neuroinclusive AI should focus on four key areas:

- **Data Infrastructure Reform:** Our study was limited by the small size of the available neurodivergent test set (220 images). Future research must prioritize the collection of large-scale, consensually annotated datasets of neurodivergent expressivity. Simultaneously, the field must move away from noisy, "scraping-based" training sets like FER-2013, which we found to induce severe calibration errors in downstream models like DeepFace.

- **Temporal and Multimodal Modeling:** Static image analysis fails to capture the dynamic nature of neurodivergent affect (e.g., stimming or temporal latency in expression). Future systems should incorporate temporal dynamics and multimodal signals (posture, prosody) to reduce reliance on rigid facial templates.

- **Participatory Design:** The definition of "ground truth" in emotion datasets remains neuro-normative. Future efforts should integrate neurodivergent individuals into the annotation process itself, ensuring that labels reflect lived emotional experience rather than external neurotypical judgment.

- **Policy and Standardization:** Finally, we recommend that regulatory frameworks categorize FER in hiring and education as "High Risk" not just due to accuracy concerns, but due to calibration failures. Mandating *Expected Calibration Error (ECE)* thresholds for specific protected subgroups would prevent the deployment of "confidently wrong" systems.

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